Question	Answer	Marks	Partial Marks
7(a)	$x = 0$	**1**	
7(b)	Tangent ruled at $x = 0.5$	**B1**	No daylight between tangent and curve at point of contact
	–9 to –6.5	**2**	**dep** on ruled tangent or close attempt at tangent at $x = 0.5$ **M1** for rise/run also dep on tangent or close attempt at tangent at $x = 0.5$
7(c)(i)	0 2.4 or better 4	**3**	**B1** for each
7(c)(ii)	Correct smooth curve	**4**	**B3FT** for 6 or 7 correct plots or **B2 FT** for 4 or 5 correct plots or **B1 FT** for 2 or 3 correct plots **FT** *their* table
7(d)	$x^3 + 3x + 4 = 10 - 8x^2$ and correctly completed	**1**	
7(e)	line $y = -2x + 2$ drawn and –0.45 to –0.35 nfww	**3**	**B2** for ruled $y = -2x + 2$ or **B1** for $-2x + 2$ seen or for line $y = -2x + c$ drawn or for $y = cx + 2$ ($c \neq 0$) drawn and **B1** for –0.45 to – 0.35 nfww
8(a)	18	**3**	**B2** for 20 nfww or **M1** for $8x + x = 180$ or better
8(b)	32	**3**	**B1** for angle $DBC = 58$ **B1** for angle $BCD = 90$
8(c)(i)	24	**2**	**B1** for angle $PRQ = 24$
8(c)(ii)	29.4 or 29.40 to 29.41	**3**	**M2** for $\frac{360-48}{360} \times 2 \times \pi \times 5.4$ or **B2** for answer (minor arc) 4.52 or 4.523 to 4.524… or **M1** for $\frac{48}{360} \times 2 \times \pi \times 5.4$
9(a)	$\frac{5}{8}$ $\frac{3}{8}$ $\frac{1}{6}$ $\frac{5}{6}$ $\frac{7}{10}$ $\frac{3}{10}$	**3**	**B1** for each pair

Question	Answer	Marks	Partial Marks
9(b)	$\frac{5}{48}$ oe	**2**	**M1FT** for *their* $\frac{5}{8} \times$ *their* $\frac{1}{6}$
9(c)	$\frac{304}{480}$ oe	**3**	**M2** for *their* $\frac{5}{8} \times$ *their* $\frac{5}{6}$ + *their* $\frac{3}{8} \times$ *their* $\frac{3}{10}$ oe or **M1** for *their* $\frac{5}{8} \times$ *their* $\frac{5}{6}$ or *their* $\frac{3}{8} \times$ *their* $\frac{3}{10}$
10(a)	75	**3**	**M2** for 79.5 ÷ 1.06 oe or **M1** for 79.5 associated with 106 [%]
10(b)	962.5 cao	**2**	**B1** for 35 or 27.5 seen
10(c)(i)	16	**1**	
10(c)(ii)	50	**1**	
10(c)(iii)	$\frac{4}{50}$ oe	**2**	**FT** *their* **(c)(ii)** for 1 or 2 marks **B1** for $\frac{4}{k}$, $k > 4$ or $\frac{k}{\text{their}50}$, $k < 50$
10(c)(iv)	19	**1**	
11(a)(i)	12.6 or 12.64 to 12.65	**3**	**M2** for $12^2 + (-4)^2$ OR **B1** for $\begin{pmatrix} 12 \\ -4 \end{pmatrix}$ **M1** for $(\text{their}12)^2 + (\text{their} - 4)^2$
11(a)(ii)	$\begin{pmatrix} -11 \\ 13 \end{pmatrix}$	**2**	**B1** for $\begin{pmatrix} -11 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 13 \end{pmatrix}$ or for $[\overrightarrow{BA} =]\begin{pmatrix} -8 \\ 7 \end{pmatrix}$
11(b)	$\frac{1}{2}$**(b – a)** oe	**2**	**M1** for correct route or correct unsimplified answer or **B1** for $\overrightarrow{QS}$ = **b** – **a** oe
11(c)(i)	$\begin{pmatrix} 9 & 50 \\ 10 & 69 \end{pmatrix}$	**2**	**B1** for 2 correct elements
11(c)(ii)	$\frac{1}{11}\begin{pmatrix} 8 & -5 \\ -1 & 2 \end{pmatrix}$ oe isw	**2**	**B1** for $k\begin{pmatrix} 8 & -5 \\ -1 & 2 \end{pmatrix}$ or $\frac{1}{11}\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ or det = 11 soi